

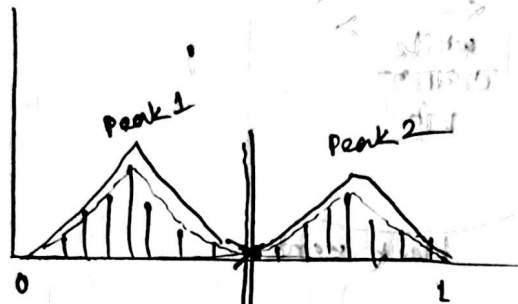
Grayscale $\leftarrow$ histogram $\rightarrow$ RGB
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

RGB 3 channel (R/G/B)
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
 possible

Segmentation

Region growing & splitting

$\rightarrow$ thresholding / threshold based segmentation.



optimal $\rightarrow$ Threshold $T1$ (let 0.6)

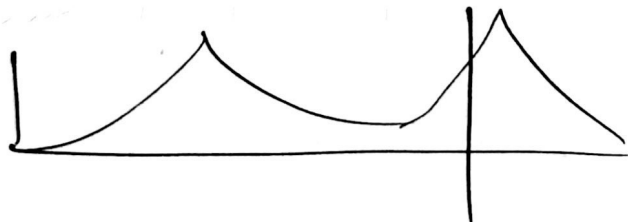
conventional approach (handcraft)

dot dot
or
cross
point

0 - 0.6 $\rightarrow$ '0' (all) black (dark).
 0.6 to 1 $\rightarrow$ '1' (all) white (bright)

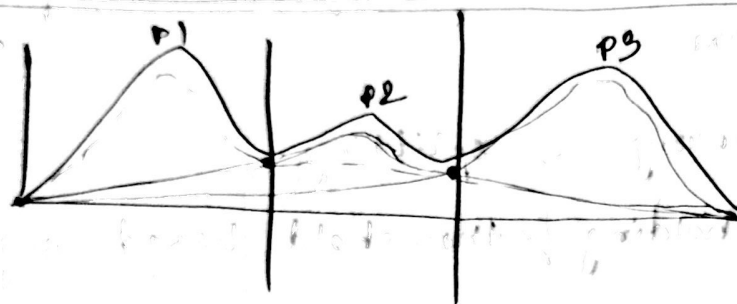


over segmentation
 under



T too large $\rightarrow$ black $\uparrow$

Double
Threshold



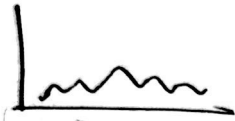
T_1

T_2

white
black

black white

local



global



$f(x, y)$

s_1

$\rightarrow$

$(f(x, y) \ominus s_1)$

$\oplus s_2$

①

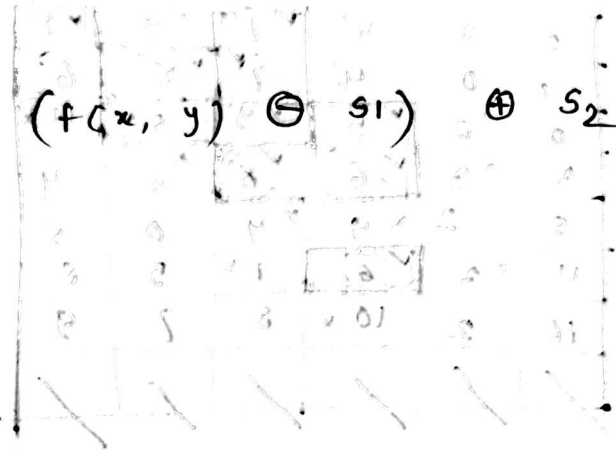
$\rightarrow$

1

□

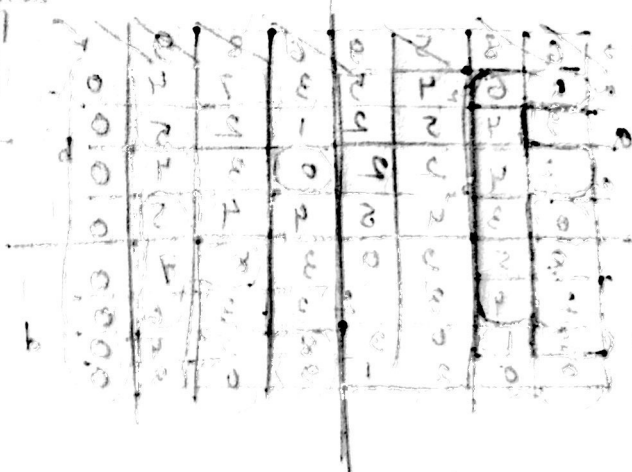
$\rightarrow$

0



pathway

$f(x, y)$



2x2 at screen 1

f(x, y) at screen 1

to screen 1

grouped at screen 1

grouped at screen 1

grouped at screen 1

grouped at screen 1

CT3

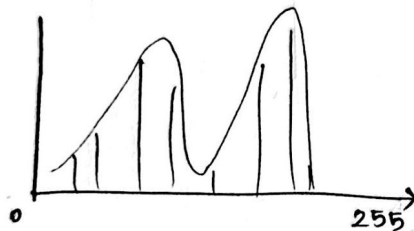
segmentation
color image processing

segmentation → partitioning into meaningful regions.

Feature → shape
size
color

Type → region based ✓
Thresholding ✓
M/L based
clustering

Thresholding



conventional image processing → mask size, operators T , given.

Grayscale → binary $\{0, 1\}$.

* binary to convert → using threshold value.

* based on threshold value:

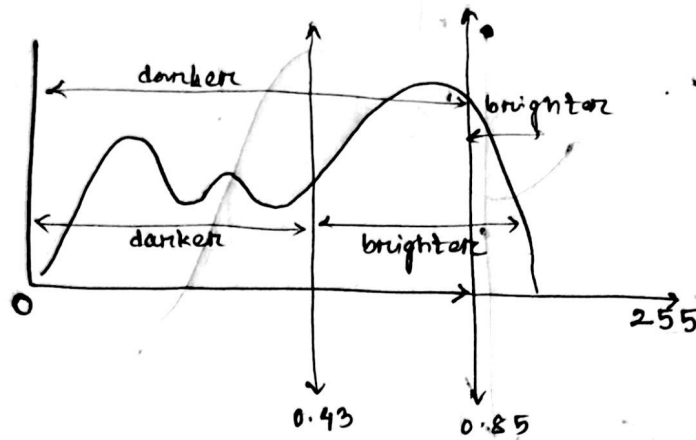
• under-segmentation → impurity Δ মোক থাকে



• over - " → যা অপ্রয়োজনীয়, deleted.



મારે problems ના રૂપ $\rightarrow$ we take optimal threshold.



$\rightarrow$ binary

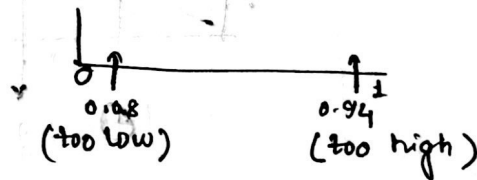
0 $\rightarrow$ dark

1 $\rightarrow$ bright

conceptual Ques.
will come.

what does
thresholding do.

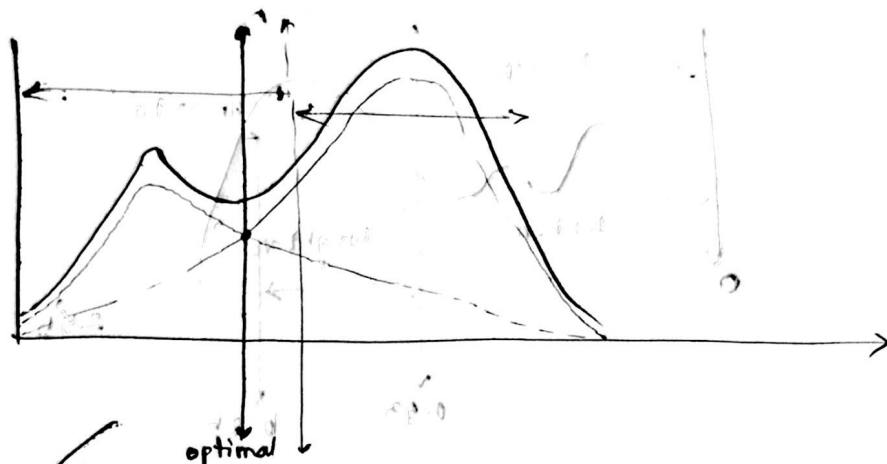
threshold low means $\rightarrow$ 0 ના ફિલ્લો (bright part તાલો)



too high $\rightarrow$ 1 $\rightarrow$ ફિલ્લો $\rightarrow$ dark part $\uparrow$.

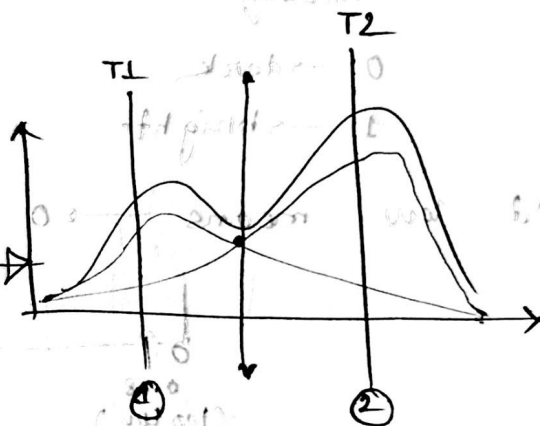
Q.1

Optimal Threshold :



Single
Multi

threshold

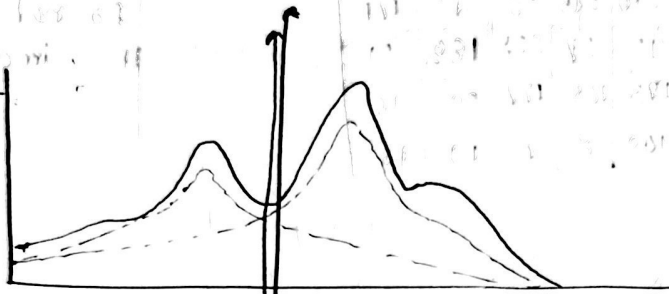


CT3

Segmentation

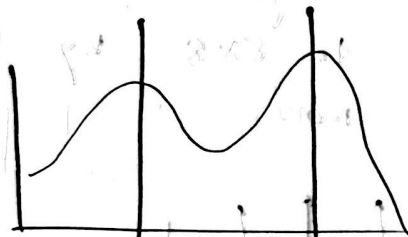
: Thresholding

Region growing / splitting



Single

Multi



splitting / merging ✓

42	100	37	102	11	30	32	21
21	23	166	179	189	168	9	5
37	180	213	48	40	230	190	3
170	51	33	210	215	39	42	171
5	190	220	42	37	225	182	21
2	3	185	173	165	177	20	16
1	100	110	109	8	7	19	15

split & merge

$$T = 20$$

Do not crop
the image block

org. image $\rightarrow P$

the size of original image block is 7×8 . For

splitting, we need an image with $2^n \times 2^n$.

So, resize the image to 8×8 by using zero
padding in the last row.

zero padding.

42	100	37	102	11	30	32	21
21	23	166	179	189	168	9	5
37	180	213	48	40	230	190	3
170	51	33	210	215	39	42	171
5	190	220	42	37	225	182	21
2	3	185	173	165	177	20	16
1	100	110	109	8	7	19	15
0	0	0	0	0	0	0	0

Region splitting.

min $\rightarrow m$
max $\rightarrow n$

intensity
diff = 1

If, $n - m > T$

split into 4 $\rightarrow s$

otherwise,

no split $\rightarrow ns$

$\alpha_L, I > T/s$

P (0/230/s)

Q (21/213/s)

b (3/230/s)

c (0/220/s)

d (0/225/s)

a_1 (21/104/s)

a_2 (57/169/s)

a_3 (57/184/s)

a_4 (33/213/s)

b_1 (11/189/s)

b_2 (5/92/s)

b_3 (59/230/s)

b_4 (9/190/s)

e_1 (2/190/s)

e_2 (42/224/s)

e_3 (0/104/s)

c_4 (9/110/s)

d_1 (37/225/s)

d_2 (6/182/s)

d_3 (0/81/s)

d_4 (0/19/ms)

Region merging :

42	100	37	102	11	30	32	21
21	23	166	179	189	168	9	5
37	184	213	48	46	230	190	3
170	51	330	216	215	39	42	171
5	190	220	42	37	223	182	21
2	3	185	173	165	177	20	16
1	100	110	109	8	7	19	15
0	0	0	0	0	0	0	0

P

